inder, the Commerce Department reported, with durable goods off 1.3% instead of the earlier-stated 1.7% But the month's ion-spending pace speeded up building outlays in all of 1971 the sharpest gain in 21 years.

(Story on Page 3)

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orm lead price of 14.5 cents a s set throughout the industry can Metal Climax and St. Joe followed Asarco to that level. imity was said to reflect a ning market that is making g less important.

(Story on Page 5)

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New Hampshire plans to build illion nuclear power complex ed Illuminating Co. and other and utilities. There would be million-kilowatt generating e completed by 1981.

(Story on Page 2)

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nmental risks of Baltimore ectric's controversial Calvert ., nuclear power plant are e," the AEC said in an imnent. The plant, cause of an of the commission's environlicies last year, was seen eavy economic benefits.

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closings have been blitzing ecently, with many employers leave the area forever. The suburban and rural regions, urring elsewhere, is said to larly strong from that city high labor costs resulting auto contracts.

(Story on Page 4)

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e time has been practically up for the second quarter, networks say, warning that tisers could miss out on the d, too. The early sellout at indicates a recovery from loss of cigaret ads and of 30 of network programming ndustry sources assert it wed business confidence.

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ed Milk Producers was the Justice Department olizing the sale of milk-profarmers in 14 states. The civil e dairy marketing coopera- the largest in the U.S., vioerman Antitrust Act.

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an company net was listed on's two leading makers. Can said profit was off e fourth quarter and 23.9% [illegible] while Continental Can [illegible] and 20. [illegible] for the sales of both were up in and year.

(Story on Page 4)

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$1.3 billion, were proposed by the President.

In a message to Congress, Nixon called for a $1 billion expansion in loans for industries and communities. He said the expansion was needed for "revitalization of the American countryside" to stem the rural-to-city population flow. The proposal would provide $345 million for commercial and industrial purposes, and $455 million to finance water and sewer systems, industrial parks and "related items which would enhance opportunities for employment," the White House said. Some 80% would be administered by the states, and 75% of the loans would be made by local banks and guaranteed by the government.

The $1 billion expansion of credit wouldn't become available until mid-1973, in an attempt to tie the loan plan to Nixon's rural revenue-sharing concept.

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A BOMB SHATTERED windows of the British Embassy in Dublin as protests continued.

An estimated 3,000 persons gathered in front of the embassy, and a police spokesman said "every available patrolman that can be spared" was at the scene trying to restrain the mob throwing bricks, bottles and gasoline bombs. Another bomb damaged the Wellington Monument. In Northern Ireland, a British soldier was gunned down in Belfast. Meanwhile, British Home Secretary Reginald Maulding said the army would continue to use "legal force" against "illegal force" in Ulster. And the British defense minister totally rejected charges that troops fired indiscriminately into crowds in Londonderry Sunday.

Ulster's Prime Minister Faulkner gave a thinly veiled warning that Protestant patience may be near an end, raising the specter of a backlash of violence.

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Italy's premier-designate failed to form a government, raising demands for early parliamentary elections. Emilio Colombo reported his failure to President Giovanni Leone as labor unions went ahead with plans for a general strike in Rome tomorrow. Observers said parliamentary elections at this time might result in new gains for neo-Fascists, and noted that the slumping Italian economy and the government's failure to restrain labor union demands were major factors in a right-wing backlash in which the neo-Fascists almost doubled their voices in local elections last June.

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Integration of Oklahoma City schools next fall was ordered by a federal judge. Noting that the case began 10 years ago, Judge Luther Bohanon called it "frustratingly interminable . . . because of the unpardonable recalcitrance" of school officials in drafting a desegregation plan. He then imposed one. In Washington, meanwhile, Attorney General Mitchell said he is opposed to merging suburban school districts with inner-city school systems just to achieve racial balance.

A move to halt job-bias bill debate was rejected, 48-37, by the Senate. The key issue is the authority the bill would give the Equal Employment Opportunities Commission to issue cease-and-desist orders to enforce its findings of job discrimination.

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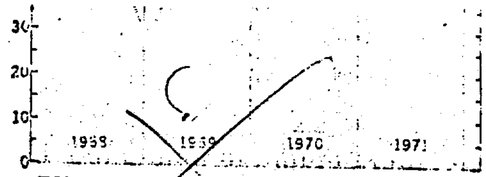
A new round of trade talks, aimed at the elimination of trade barriers with the U.S. and as many other countries as possible, will be proposed by the six Common Market foreign ministers, sources said. The proposal is to be



BUSINESS FAILURES fell to a seasonally adjusted rate of 37.5 per 10,000 in December from 41.6 in November, Dun & Bradstreet Inc. reports.

Be Kind to Plants— Or You Could Cause A Violet to Shrink

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How One Lie-Detector Expert, In Turning Over a New Leaf, Decided Plants Have Feelings

By RICHARD MARTIN

Staff Reporter of THE WALL STREET JOURNAL

NEW YORK—Have you heard about the little old grandma who swears that by whispering words of encouragement to her geraniums and African violets she helps them grow?

Cleve Backster has — many, many times. Wherever his travels take him, whether it's Haiti, Lebanon or Grundy Center, Iowa, he runs into people just like her.

Mr. Backster isn't a plant fancier; he's a polygraph expert, and he doesn't talk much to plants himself. But he does study them a lot, and his observations have convinced him that while plants don't exactly listen to Granny's sweet talk, they do indeed get the message—probably by reading her mind.

His experiments, in fact, seem to indicate that besides some sort of telepathic communication system plants also possess something closely akin to feelings or emotions, says Mr. Backster. They appreciate being watered. They worry when a dog comes near. They faint when violence threatens their own well-being. And they sympathize when harm comes to animals and insects close to them.

How It All Started

Mr. Backster got hooked on plant studies Feb. 2, 1966, in the interrogation room of the Backster School here where he trains private investigators, police and government personnel to use polygraph machines—lie-detectors. He wondered how long it would take water he had just given a tall, droopy-leafed dracena plant to travel from the roots to the leaves, so he connected a pair of polygraph electrodes to a leaf, figuring that the moisture might gradually change its resistance level enough to register on his lie-detector.

To his surprise, he got an immediate polygraph reaction pattern that closely resembled that of a person under emotional stimulation. Wondering whether the plant would also produce a reaction similar to a human's if its safety were threatened, Mr. Backster decided to try burning a leaf. But before he could reach for a match, "at the split second that I had the image of fire in my mind, the rec[illegible] [illegible] right off the top of the chart," he recalls. "It really shook me up."

Since then, Mr. Backster estimates he made "several thousand observations [illegible]" [illegible]

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Gulf Sulphur raised its prices low mineral in the Southern ere competition from low- ports has triggered dumping by Freeport Minerals Co. Mexican producers. Sources the unusual price increase Gulf's having its own Mexi- ction, although it denies that ing. The U.S. industry also hit by a glut of cheap sulphur ada.

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proval of a far-reaching cos- ustry proposal for an experi- self-regulation is expected pite consumer groups' pro- companies would voluntarily nd disclose secret formulas, he agency to spot hazards dangerous products off the But critics say that would ctive public protection.

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ia Gas agreed to have Apco own and operate for it a $30 ant to make synthetic gas tha that New England Petro- d provide from its Bahamas ith California Standard.

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Steel's mill in Fontana, struck by about 6,500 pro- d maintenance workers in a er a plan for employes to roduction cost savings, the ustries unit said.

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w Jones 40 bonds 74.12, up 0.06. es — Dow Jones futures index ; spot index 148.10, up 0.03.

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	16	Securities Mkts.	23-28
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A move to halt job-bias bill debate was rejected, 48-37, by the Senate. The key issue is the authority the bill would give the Equal Employment Opportunities Commission to issue cease-and-desist orders to enforce its findings of job discrimination.

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A new round of trade talks, aimed at the elimination of trade barriers with the U.S. and as many other countries as possible, will be proposed by the six Common Market foreign ministers, sources said. The proposal is to be made tomorrow, and will include offers to help the U.S. sell some wheat, oranges and grapefruit stocks. One source described the new round of talks as being comparable to the Kennedy Round, which led to agreements on reciprocal reductions in tariffs.

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Saigon fighter planes joined U.S. warplanes in bombing Communist supply trails in Laos, the Saigon command said. The announcement, the first official word the South Vietnamese were attacking by air into Laos, said the jets have been flying missions against the Ho Chi Minh trail since Dec. 1. In the ground war, two U.S. soldiers were killed and four wounded in three battles east of Saigon.

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Daniel Schorr expressed skepticism over reports he was being considered last fall for a post with the Council on Environmental Quality. Schorr noted an FBI investigation of him, allegedly in connection with the job, was begun after his CBS reporting brought complaints from Nixon, GOP National Chairman Rep. Dole, and White House staffers. If the White House was considering him for a job under those circumstances, Schorr told a Senate committee, "it would have been an extraordinarily open-minded thing for them to have done."

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Nixon and Sen. Kennedy were listed on Wisconsin's presidential primary ballot, but a Kennedy aide said the Massachusetts Democrat would ask that his name be withdrawn. They were among 16 politicians assigned spaces on the ballot. In Washington, meanwhile, Commerce Secretary Stans said Democratic contentions that Nixon has already amassed a $50 million warchest were "an outright lie." He said the Democrats were hoping to discourage contributions to the GOP, which presently, he said, has a $2 million treasury.

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A photo album of Howard Hughes' aides was assembled by federal and Manhattan prosecutors. They will ask novelist Clifford Irving to try to identify one of the aides as the "George Gordon Holmes" he claimed served as a go-between during the writing of the alleged Hughes "autobiography." One former aide, John Meier, already has denied reports that he was the man who dealt with Irving. Meier is a candidate for New Mexico's Democratic senatorial nomination.

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Twenty-two major paintings, including works by Cezanne and Picasso, were donated to the National Gallery of Art by the W. Averell Harriman Foundation. The gift is in memory of Marie N. Harriman, late wife of the former ambassador.

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Died: Howard Barlow, 80, symphony conductor, of a heart attack in Bethel, Conn.

figuring that the moisture might gradually change its resistance level enough to register on his lie-detector.

To his surprise he got an immediate polygraph reaction pattern that closely resembled that of a person under emotional stimulation. Wondering whether the plant would also produce a reaction similar to a human's if its safety were threatened, Mr. Packster decided to try burning a leaf. But before he could reach for a match, "at the split second that I had the image of fire in my mind, the recording pen bounded right off the top of the chart," he recalls. "It really shook me up."

Since then, Mr. Backster estimates he has made "several thousand observations, mostly pretty carefully done, but including a few sloppy ones, too," in his efforts to find evidence of perception capabilities in house plants, fresh fruits and vegetables, fresh eggs, mold cultures and human blood and tissue samples. Some of his observations have been as surprising as the first one.

When the Killer Appears . . .

Some samples:

—Polygraph electrodes are attached to three fresh vegetables and someone picks one and drops it into boiling water; the one selected seems to "faint" before it is even touched (that is, it registers on the polygraph chart a sudden upward bound followed abruptly by a straight line), but the other plants don't react (their polygraph squiggles continue uninterrupted). Eggs similarly "faint" when they are about to be picked up and broken, and they also get nervous when another egg is broken nearby.

—Six students draw lots to see who will uproot and tear to shreds one of two plants alone in a room. Later, the surviving plant shows no reaction when five of the students reenter the room, one by one, but it faints when the plant killer returns.

—An automated device rigged up in one room randomly selects and dumps cups of tiny brine shrimp into boiling water; meanwhile, plants being polygraphed in another room react at the instant the brine shrimp hit the water and die.

—An egg, fresh from a nearby grocery store, is polygraphed and shows pulsations that closely coincide to the heartbeat of a chick embryo; but under microscopic examination, the content of the egg shows no evidence of the beginnings of a physical or circulatory structure that could account for the pulsations.

"These are startling findings," says William L. Bondurant, top executive of the Mary Reynolds Babcock Foundation in Winston Salem, N.C., which has just given Mr. Backster $10,000 to help further his research.

"It's a rather unusual grant for us," adds Mr. Bondurant; mostly the foundation's funds go to support Southeastern colleges and universities. "It's risky research, certainly. But his work indicates that there may be a primary form of instantaneous communication among all living things that transcends the physical laws we know now—and that seems to warrant looking into."

Mr. Backster started looking into it without benefit of academic credentials. He studied civil engineering, agriculture and psychology at Texas University, Texas A&M and Middlebury College in Vermont, but he still lacks about one semester's credits for a bachelor's degree.

He got into lie detection work during and after World War II, first as a Navy ensign in the South Pacific, then as an Army master ser-

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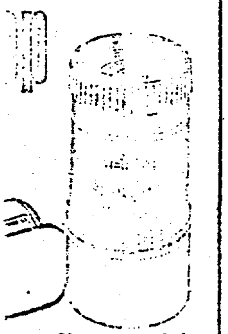

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King C. Gillette, a traveling salesman, invented the safety razor. In recent decades, it

Be Kind to Your Plants—Else You Could Cause a Violet to Shrink

Continued From Page One

geant at Fort Holabird, Md., and later as a polygraph section head for the Central Intelligence Agency. He opened the Backster School in 1951. It trains about 50 polygraph operators a year.

He also heads Backster Associates, a polygraph consulting and personnel screening service, and Backster Research Foundation, his nonprofit plant-studies operation. All three concerns are located in a warren of cramped, cluttered offices and stark classrooms in a slightly seedy office building a few blocks from Times Square.

Mr. Backster lives two blocks away in a sparsely furnished two-bedroom apartment that's devoid of plants. The 47-year-old bachelor spends most of his time these days polygraphing plants and trying to interest others in his work; a three-man staff does most of the teaching and consulting work.

In the past year or so, Mr. Backster has talked to over 100 different groups, including Yale linguistics students, Dartmouth biologists, Newark College of Engineering alumni, International Nickel Co. technicians, theology students at several universities and researchers at the Naval Weapons Center in White Oak, Md.

Turning on Scientists

"I don't like to spend a lot of time talking to groups interested in gardens or psychic phenomena," he says. "I'm much more interested in getting scientists turned on to proving these things by replicating my work." He claims he's already in touch with scientists doing similar research using polygraphs, electrocardiograms and similar equipment at "25 or 30" universities, but he declines to name any. "You'll have to wait until they're ready to publish," he says.

Among some of those familiar with Mr. Backster's work, reactions are mixed. Professor Bernard Grad, who is doing biological studies on aging in the psychiatry department at McGill University, Montreal, says cautiously: "It's very intriguing, but I'm not an expert in polygraph work, so I don't want to overplay or underplay it."

A spokesman for Bell Telephone Laboratories' Illinois facilities, where Mr. Backster last year spoke to a seminar of supervisors, will say only that he gave "an entertaining presentation." But a Bell Labs man who heard Mr. Backster speak to an earlier seminar of New Jersey Bell Labs engineers and technicians says: "He was a tremendous hit and everything he said seemed absolutely credible."

"It's the sort of thing some people would dismiss out of hand," says Arthur Pinkerton, a 38-year-old physicist whose specialty is cancer radiation therapy. He met Mr. Backster while on the staff at New York's Sloan-Kettering Cancer Institute, and "I fooled around with that stuff myself for awhile," he says. "When you look into it you come away with the feeling that there is certainly something going on."

Luring Serious Researchers

By sticking to observing plants and eggs and making soft-sell speeches to lots of scientific groups, Mr. Backster figures he stands the best chance of luring serious re-search this way," he explains. "Either they see the implications of my observations and become intensely interested or they just ignore me completely."

The pulsating egg "could have profound implications for origin-of-life research," he thinks. "This gets us into a force field that may regulate development of the circulatory system, some sort of genetic blueprint that hasn't been known to exist."

Mr. Backster isn't quite sure himself just what all the implications of his observations might be. "We can test fruits and vegetables until they are cooked or completely rotted away and still get reactions," he says. "This capability seems to extend right down to the single cell level." Furthermore, he says, the signals can't be screened out by distance, lead shields or the electronic fields of interference that interrupt conventional communications.

"We're getting into another dimension, a scientific twilight area in which something can go from point to point without going between them and without consuming time to get there," he says. "But that won't be as weird as we might think; it could simply mean that some of the things that were once laughed at in theoretical physics are finally falling into place."

"I'm also beginning to see a lot of similarities between the vehicle involved here and the phenomena of prayer," says Mr. Backster. "As a former agnostic who didn't take the trouble to be an atheist, I see some very high-level theological and spiritual implications of this—it opens the way for scientific exploration of the concept of the soul and seems to indicate a scientific justification for the power of prayer."

An Unhappy Experience

He also sees "fantastic implications in this for the science of crime detection." But Mr. Backster says he's leaving those for others to explore, since he's already had one unhappy experience in that area.

He tried a variation of the "plant killer" experiment in an actual police case more than a year ago. A girl had been murdered late at night in a factory office and scores of maintenance personnel were among the suspects. Flabbergasted police officers let Mr. Backster polygraph two plants that had "witnessed" the crime as the plant workers paraded through the office one by one. The plants failed to react and conventional police investigative procedures later turned up the killer outside the company.

Word of the experiment leaked out in a National Wildlife Magazine story about Mr. Backster's plants recently. ("I thought we were off the record," he says.) When they heard about the story, other members of the polygraph operators' fraternity threatened to expel Mr. Backster from their national association if he ever tried such a thing again.

So he's gone ahead [illegible] other things, including [illegible] he thinks will [illegible] that plants possess the ability to remember. He's also doing more work with eggs, including trying to find out whether those from the same hen react to each other the same way those from different hens do. And, he says, "I want

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